

ATmega64 Morse Generator

Timer Configuration and Timing Calculations

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1. Clock Configuration

Microcontroller:

ATmega64

Clock Frequency:

$F_{CPU} = 8 \text{ MHz}$

Selected Timer:

Timer0

Prescaler:

1024

2. Timer Tick Calculation

Timer Tick Duration:

$\text{Timer Tick} = \text{Prescaler} / F_{CPU} = 1024000 / 1024 = 1000 \text{ sec}$

3. Required Delay Calculation

Desired base delay:

25 ms

Required timer counts:

Counts = $25\text{ms}/1\text{ms} = 25$

4. Timer Initial Value

Timer range:

0–255

Initial value:

$256 - 25 = 231$

TCNT0 = 231

5. Timing Relationships

Dot duration:

25 ms

Dash duration:

75 ms

Inter-element delay:

25 ms

Character delay:

75 ms

Word delay:

200 ms

6. Timing Ratios

Dash = 3 × Dot

Character Gap = 3 × Dot

Word Gap ≈ 8 × Dot

7. User Timing Modes

Mode 1:

Dot = 25 ms

Dash = 75 ms

Word Gap = 200 ms

Mode 2:

Dot =50 ms

Dash =150 ms

Word Gap =400 ms

Mode 3:

Dot =75 ms

Dash =225 ms

Word Gap =600 ms

Mode 4:

Dot =100 ms

Dash =300 ms

Word Gap =800 ms

Additional modes continue similarly.

8. Design Considerations

Fixed timing ratios were maintained because changing ratios dynamically would:

- Increase implementation complexity
- Increase resource usage
- Reduce Morse readability
- Create waveform inconsistencies